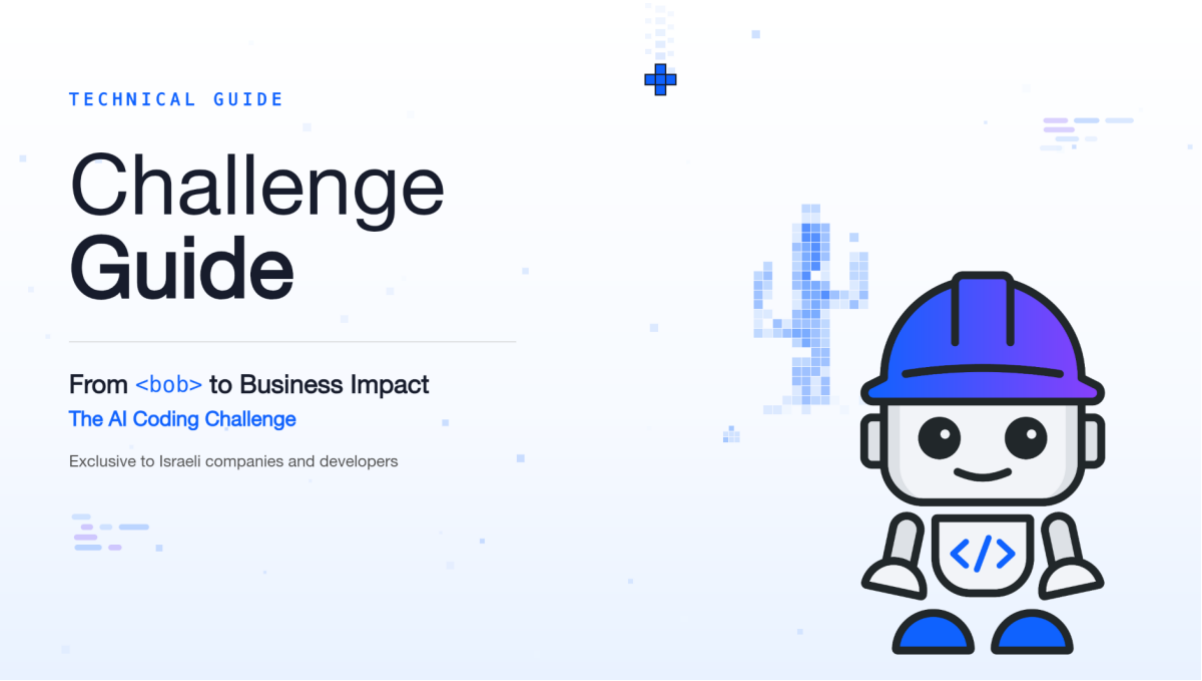


Technical guide for participants

Your hackathon accounts, IBM Bob, and watsonx Orchestrate — set up and connected.

Version	Audience
3.0	Registered participants

SUPER TIP

Upload this document into Bob to ask any question about the guide.

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1 Introduction

This challenge is built around IBM Bob — an AI coding assistant that works alongside you in your codebase. Everything here exists to get you from a clean machine to building, on the accounts provisioned for the hackathon.

1.1 What you are going to do during this challenge

You have built with AI. Now see how far you can take it.

This is more than a coding challenge. You bring a real problem — from your organisation, or from your own needs — build a working solution for it, and then make the case for that solution inside your company or for your own business. For the strongest teams it ends in a pitch to senior executives from Israel's tech industry.

It runs in five stages:

Stage		What happens
01	Choose	Start with something worth solving — an enterprise challenge, a problem in your own business, or an idea you want to put to the test.
02	Build	Turn the idea into a working proof of concept with IBM Bob and AI agents, with access to IBM experts along the way.
03	Make the case	Turn the build into a business proposition — the problem, the business value, and the reason to take it forward, whether that is inside a company or as your own venture.
04	Pitch	Learn how to explain what you built, why it matters, and what should happen next.
05	Make the cut	The strongest solutions move to the final stage, with the opportunity to pitch to senior executives from Israel's tech industry.

Your idea → Working solution → Business case → Management pitch → Final

NOTE

This guide covers stage 02 — the build. Stages 03 and 04 are supported separately: teams who want help shaping the business case and preparing the pitch can join the Spyre Group workshops and office hours.

IMPORTANT

One technical thing to keep in mind from day one: judging looks at the Bob task session report, exported from the IDE. That means the build has to happen inside Bob IDE on your hackathon account — not on a personal account, and not outside the tool.

NOTE

The hackathon site is the authority on rules, deadlines and submission format. This guide covers the technical setup; where the two ever disagree, the site wins.

1.2 The challenge is yours

There is no fixed brief. Pick the thing that would actually create value — where you work, or in your own business. In practice, most teams go in one of these directions:

- **Create a new application** — from blank page to something that runs.
- **Modernise an existing one** — take something legacy and bring it forward.
- **Integrate Bob with MCPs** — connect Bob to your own systems and work through them.
- **Anything else that brings value** — if you can argue the impact, it qualifies.

TIP

The judges are looking for business impact, not novelty. A modest idea that genuinely saves real time — for a department, a small business, or you — will beat an ambitious one that does not run.

NOTE

Stuck for an idea? Do not brainstorm alone. We strongly recommend two conversations: one with the business people around you — in your company, among your clients, or whoever actually lives with the problem — they will hand you a use case in ten minutes. And one with us: we see what other teams are attempting and what the tools do well, and we are happy to think it through with you.

1.3 Bob covers the whole process, not just the code

IBM Bob is IBM's enterprise AI coding environment, built to help teams develop AI solutions with the security, governance and integration capabilities organisations actually need.

Most people underuse it by treating it as autocomplete. It can carry the work from the first sketch to the deployed application:

Stage	What Bob can do
Design	Write the design paper, the blueprint and the specification with you

Architecture	Produce the architecture document
Design tooling	Connect to your other design tools for the design phase
Build	Write the application, with a deep set of features to support you

1.4 Two accounts, both provisioned for the hackathon

You will be given access to two separate things. They arrive differently, so it is worth knowing which is which before you start.

Account	How you get it	Needed?
IBM Bob	Email invite, sent automatically to every registered participant at the start	Required
IBM Cloud (watsonx Orchestrate)	Requested by your team from the hackathon site, then an email invite per member	Optional — if you are building agents

IMPORTANT

If you already use Bob on a personal account, use the hackathon-provisioned account only for this event. Work done on your personal account consumes your own usage, and will not count for the hackathon.

NOTE

An IBMid is required to authenticate with both Bob and IBM Cloud. If you do not have one, create it for the email address you registered with, before the challenge opens.

1.5 You are not doing this alone

On the first day of the challenge an IBM AI engineer is assigned to you. They are yours for the whole challenge — contact them whenever you need technical help, whether that is an architecture question, something broken in the setup, or a use case you are not sure how to approach. There is no quota and no need to save it for emergencies; teams that use their engineer early get further.

1.6 At a glance

Challenge	From <bob> to Business Impact — The AI Coding Challenge

Who it is for	Developers, architects, AI engineers, tech leads and technology teams — enterprises, small businesses, startups and independents, in Israel. Business stakeholders are also invited to join teams to help define business needs, validate ideas, and provide domain expertise.
What to bring	An idea, a technical challenge, or a real use case — from your organisation or from your own needs
Registration	https://compete.092601.watsonx-challenge.ibm.com/competitions/bobisraelhack
Details and webinar	https://bluesolutions.co.il/ibm-bob-challenge-2026/
Bob education	https://bob.ibm.com/docs/ide

2 Part 1 — Your hackathon Bob account

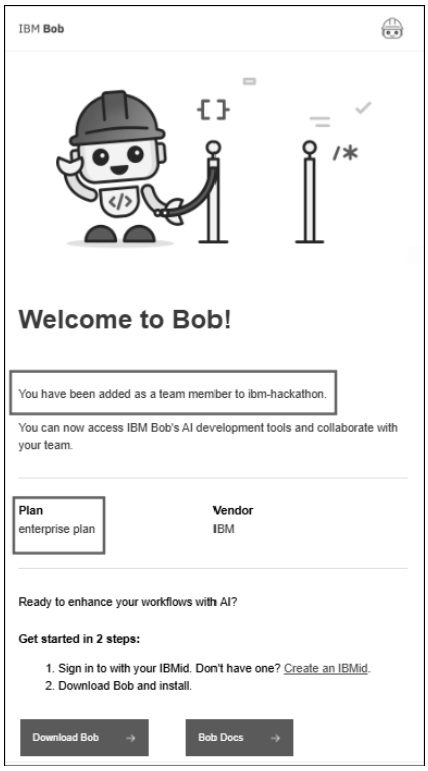
Access is tied to your hackathon registration. Everything below assumes you registered with the email address you are now using.

2.1 Confirm your hackathon Bob account

At the start of the hackathon, every registered participant receives an email invite from the IBM Bob team. Open the inbox you registered with and look for it.

You have the right email if it says:

- You have been added as a team member to `ibm-hackathon-xxxx`
- Bob Plan: **Enterprise plan**



The hackathon invite. Note the team line and the enterprise plan — and that it tells you to sign in with your IBMid, then download Bob.

TIP

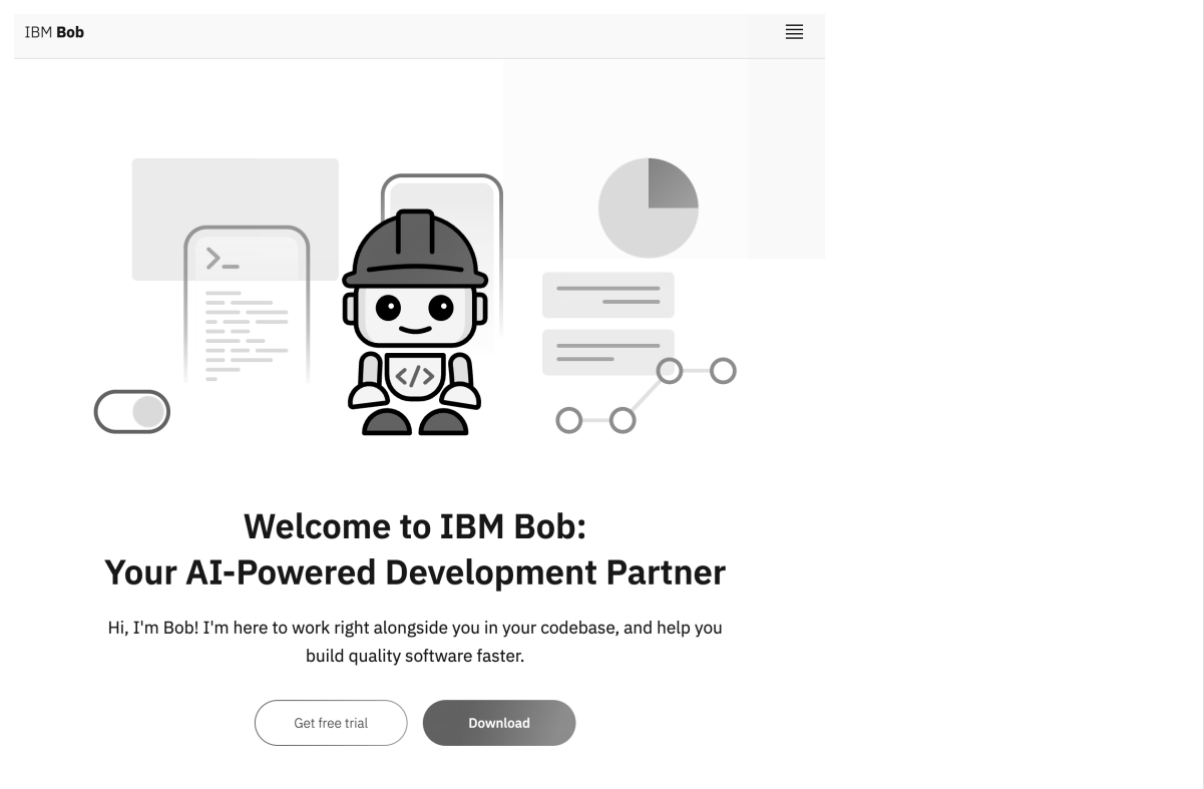
Cannot find it? Check Junk and Spam, and search your mailbox for “IBM Bob”.

NOTE

“Team member” here is account terminology, not a hackathon team. It is simply how access is granted through an Enterprise account. If you want to join or form a team, do that through the hackathon platform.

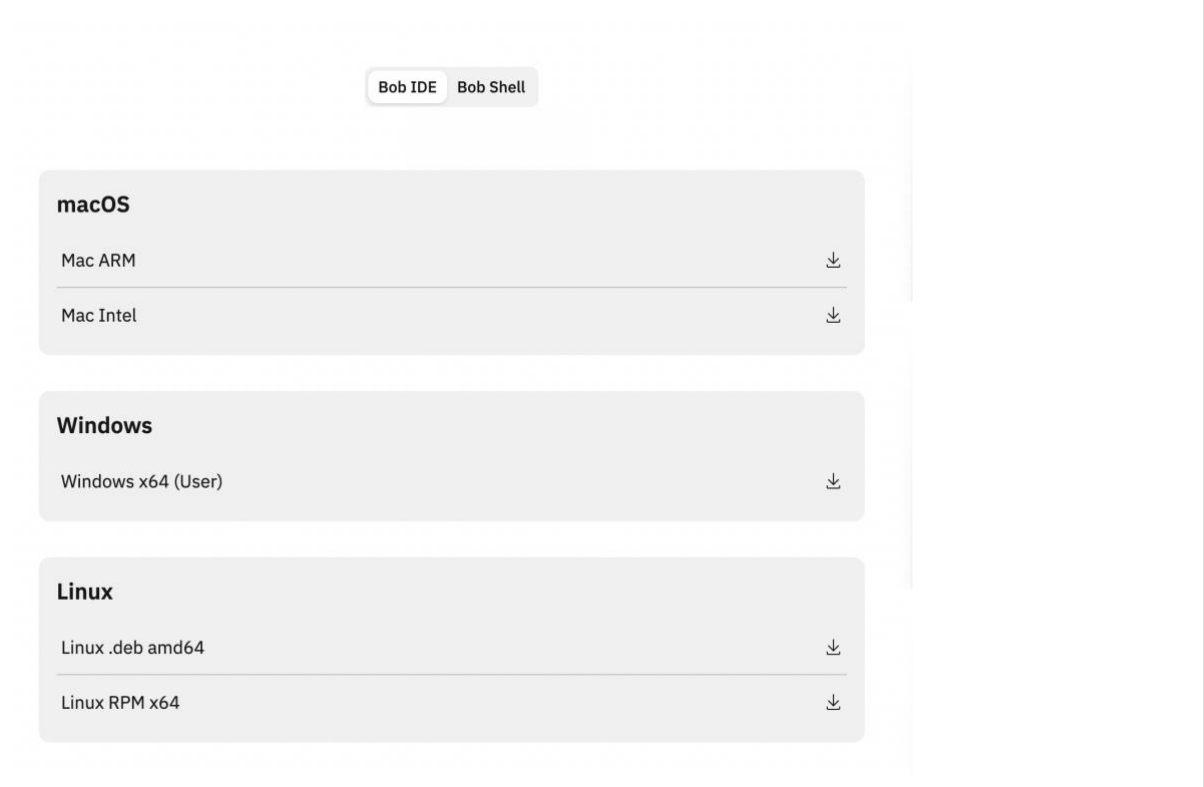
2.2 Install Bob IDE

1 Open the Bob site and choose Download.



The Bob welcome page. Download is the blue button.

2 Pick the build for your machine — macOS (Mac ARM or Mac Intel), Windows x64, or Linux (.deb or RPM). **Take Bob IDE, not Bob Shell.**



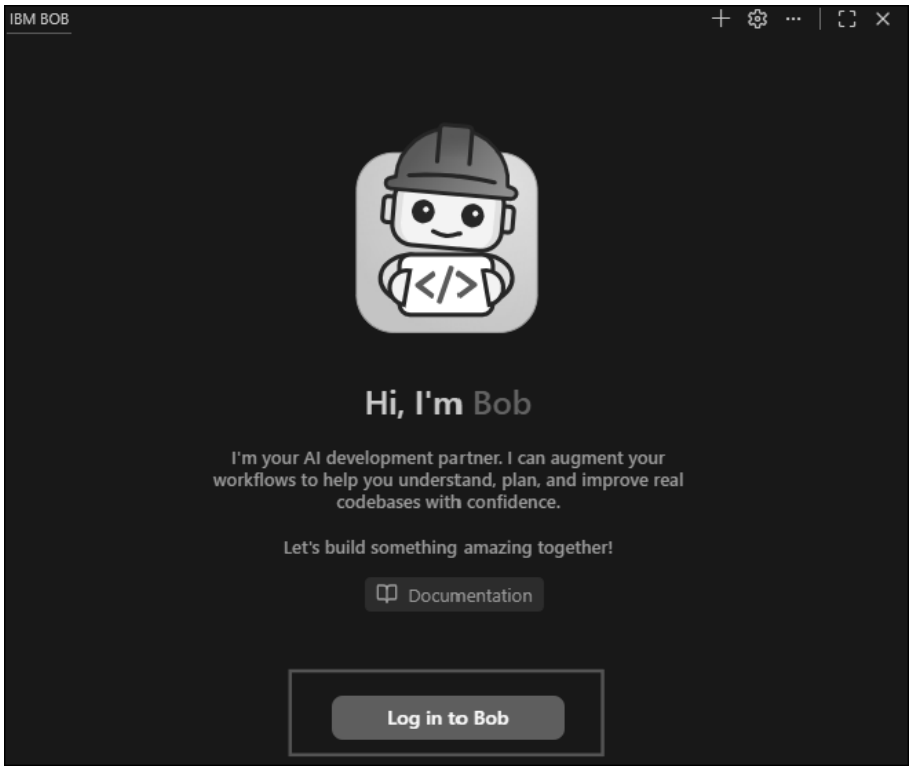
Download options. Mac ARM and Mac Intel are not interchangeable — check which machine you are on.

IMPORTANT

Bob IDE is required for this hackathon. You will need to export the Bob task session report for judging, and that export comes from the IDE. Bob Shell is optional and can be used alongside it, but not instead of it.

2.3 Sign in to Bob

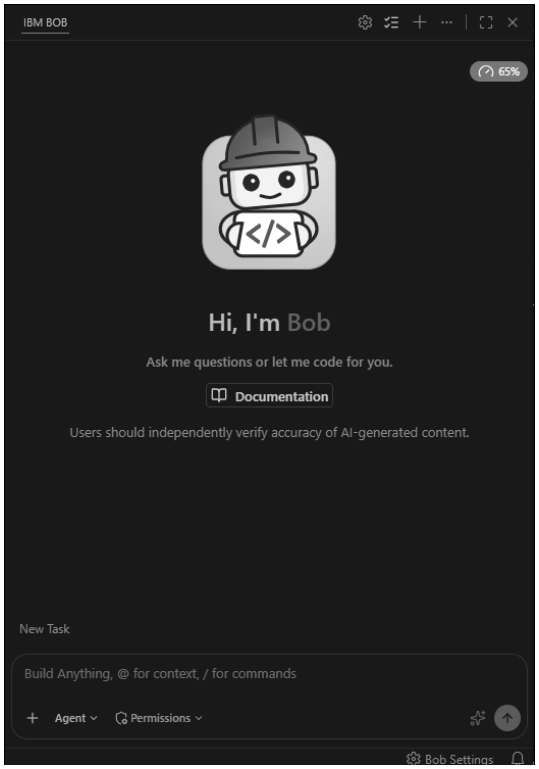
1 Open Bob IDE and select **Log in to Bob**.



Log in to Bob sits at the bottom of the welcome panel.

2 Complete the IBMid authentication in your browser, **using your hackathon registration email**.

3 When it finishes, the Bob chat interface opens. That is your workspace — you are ready.



Signed in and ready. The usage indicator top right tracks your Bobcoins as you work.

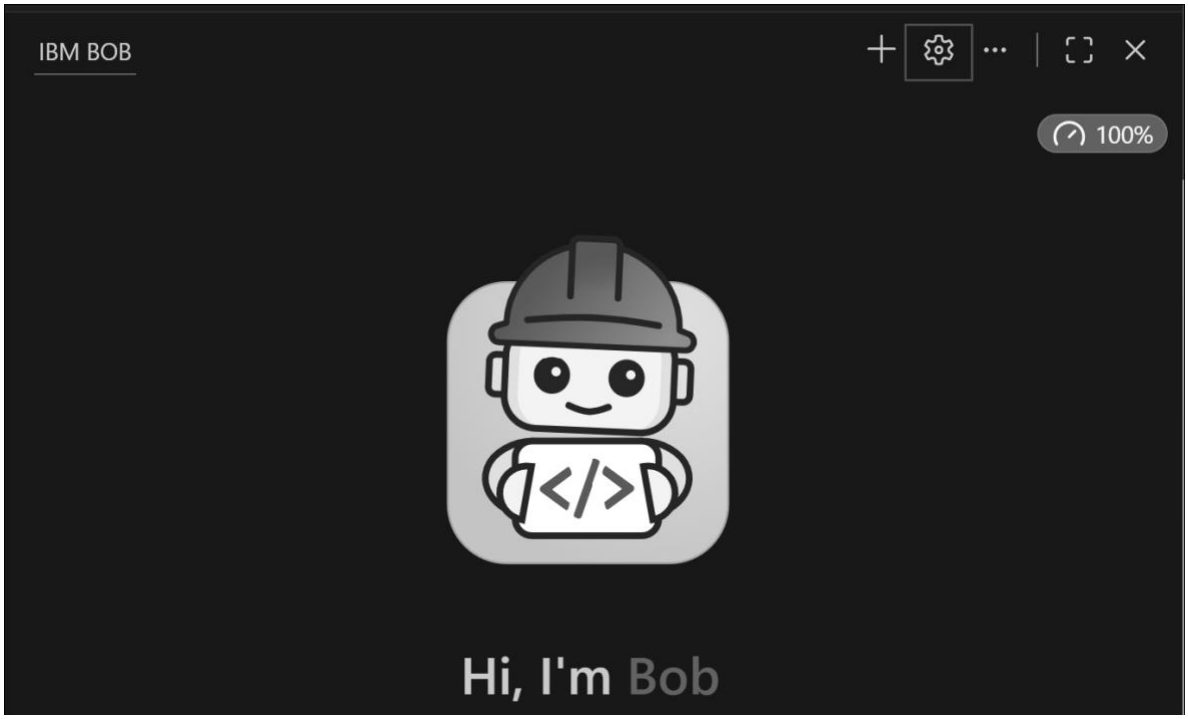
NOTE

If outbound traffic is blocked on your network, you may need to adjust your firewall rules before Bob can authenticate.

2.4 Check you are on the hackathon account

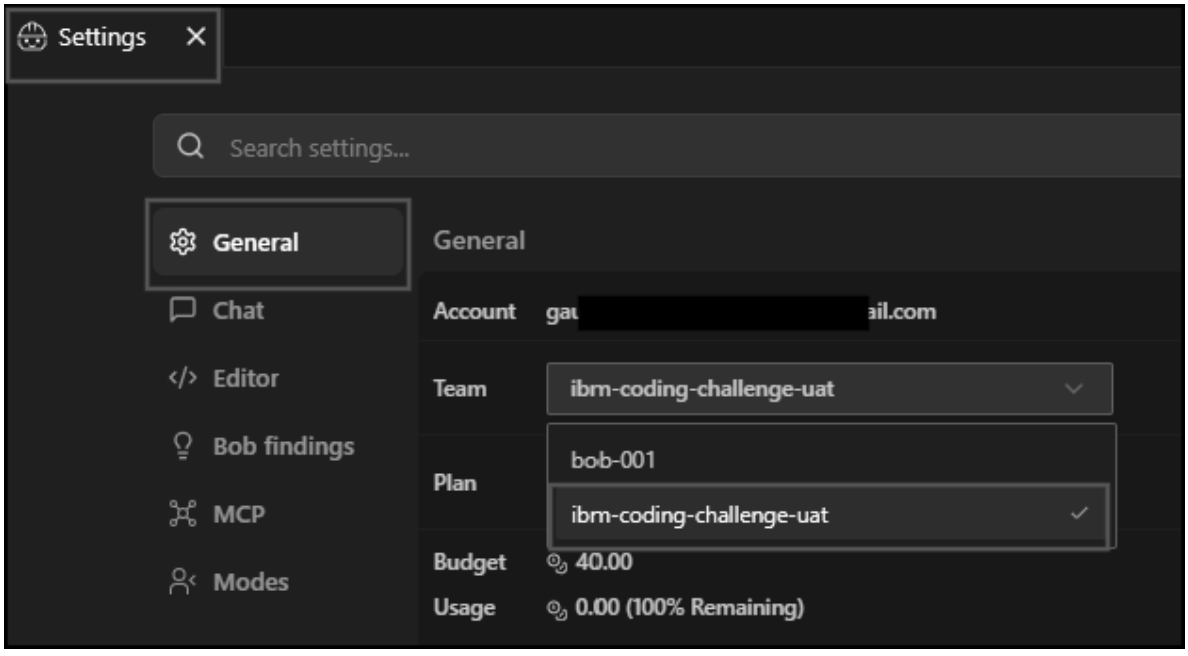
If you have access to more than one Bob account, Bob may sign you into the wrong one. Ensure that you select the hackathon provisioned instance named **ibm-coding-challenge-xxx (region: us-east)**.

1. Open Settings from the Bob header.



The settings icon sits in the Bob header.

2 Under **General**, open the **Team** selector and switch to the hackathon instance — `ibm-coding-challenge-uat` (region: us-east). Anything else is a personal account.



The Team dropdown. Pick the hackathon instance, not your personal one.

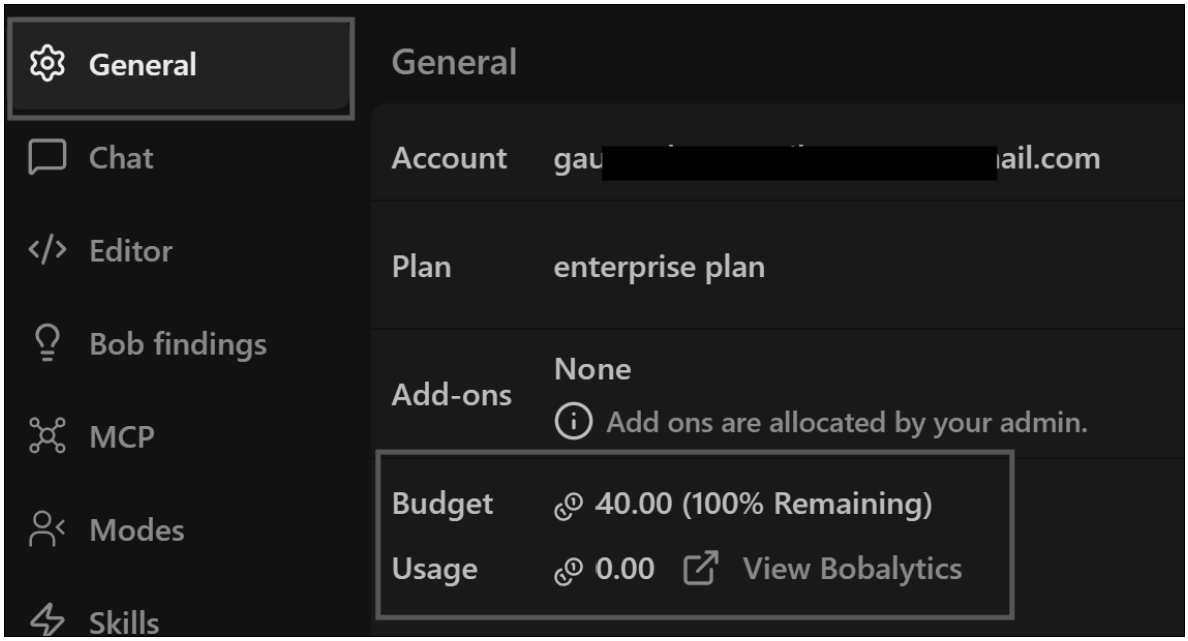
NOTE

As with the invite email, the team name here is account terminology and does not represent your hackathon team.

2.5 Your Bobcoins

Every interaction with Bob's AI capabilities consumes Bobcoins.

You can see the balance in the same place you just checked the team: **Settings** → **General**. Budget shows the allocation and what remains; Usage shows what you have spent.



TIP

Divide tasks with your teammates deliberately. Each member has their own allocation, so the team total is what you actually have to spend — but only if the work is spread across accounts.

Run out of Bobcoins?

If you are running low on or have run out of Bobcoins, contact the IBM hackathon team for assistance. Requests for additional Bobcoins will be reviewed and supported on a case-by-case basis.

3 Part 2 — watsonx Orchestrate (optional)

NOTE

This whole part is optional. You only need watsonx Orchestrate if you are building an AI agent. If your solution is an application rather than an agent, you can stop after Part 1 and start building.

If your use case does involve agents, we strongly recommend using watsonx Orchestrate. The integration with Bob is unusually simple and natural — and it is also where you can keep going if your Bobcoins run low.

3.1 Request your team's IBM Cloud account

Unlike Bob, this account is not sent to you automatically. Someone on your team has to ask for it.

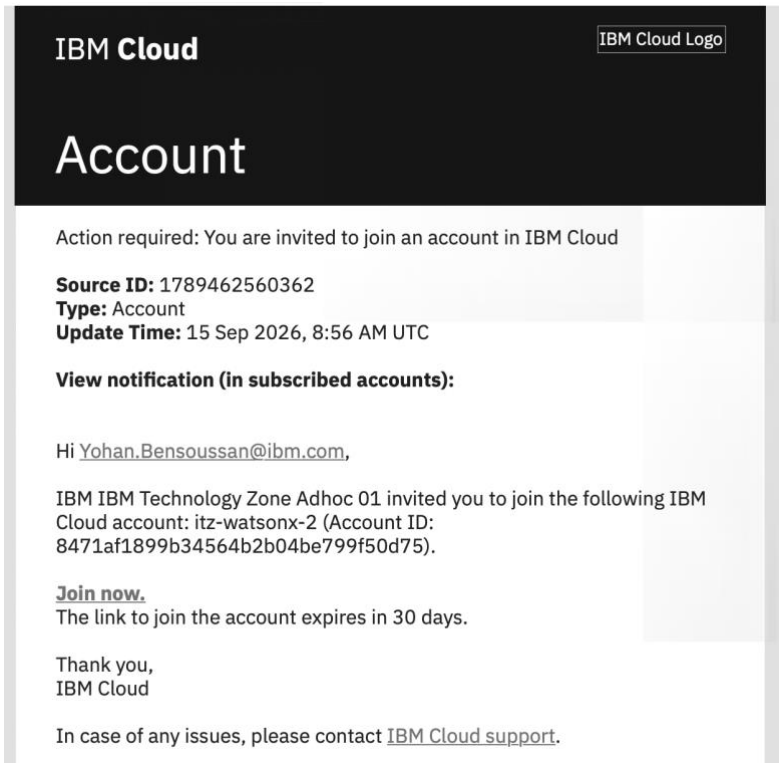
1. Sign in to the hackathon site.
2. Go to the **"Complete the hackathon"** page and follow the instructions there to request a pre-configured IBM Cloud account for your team.
3. Once it is provisioned, every member of the team receives an email invite.

TIP

Do this on day one. The account is per team, not per person, and nothing in Part 2 can start until it exists.

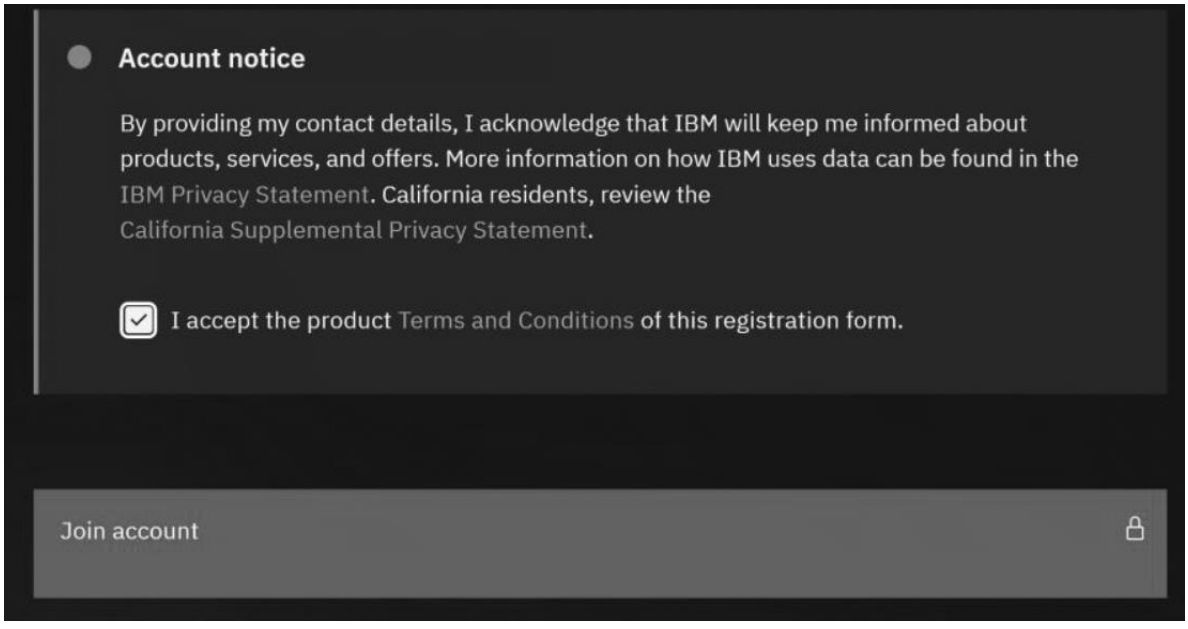
3.2 Join the account

1. Find the IBM Cloud email in the inbox you registered with and click **Join Now**. Check Junk and Spam, or search for "IBM Cloud".



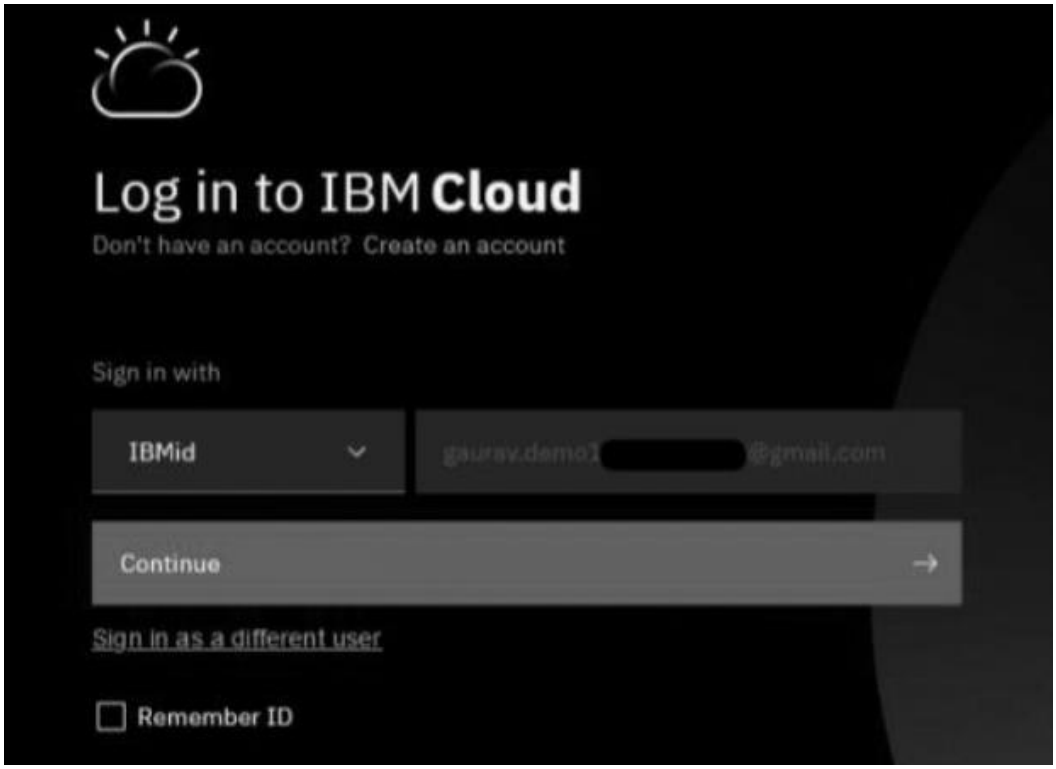
The invitation email. The join link expires after 30 days.

2 Review your account and personal information, read and accept the Account notice, then click **Join Account**.



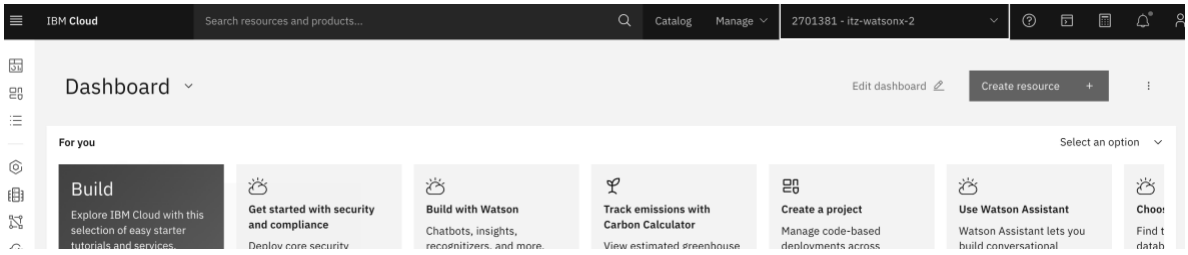
Accept the terms, then Join account.

3 Complete authentication by clicking **Continue**.



IBMid authentication.

4 You land on the IBM Cloud dashboard.



The account selector sits in the top bar, beside Manage.

IMPORTANT

If you already have a personal IBM Cloud account on the same email or IBMid, you may be dropped into that one instead. Open the account dropdown at the top right and switch to the hackathon `watsonx` account before you do anything else.

3.3 What is in the account — and what is not

The account is pre-configured with only what the hackathon needs. You cannot add services or change permissions, and that is deliberate — if something looks locked, it is not part of the hackathon.

Also available (optional)	Speech-to-Text · Text-to-Speech · Cloudant

Out of scope	AgentOps (Preview)
Your role	Builder — assigned by default and cannot be changed

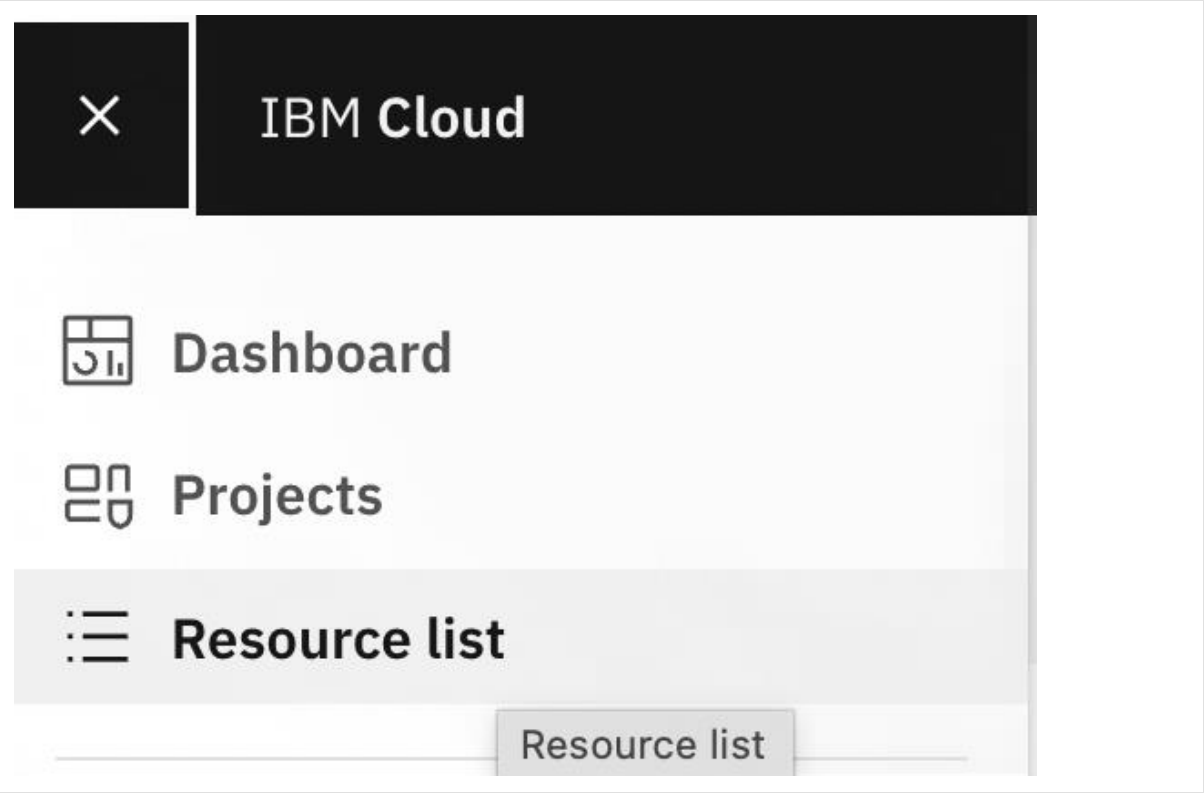
3.4 Protect your credentials

> **IMPORTANT**
>
> **Never commit an IBM Cloud API key to a public repository.** If a credential from your hackathon account is found on a public platform it is deactivated immediately and your cloud account is suspended until you remove it from all public sources, rotate it, and confirm remediation with the hackathon support team. This is the fastest way to lose a day of the challenge.

It happens most often at submission time, when a working project is pushed to a public repo for judging. Check before you push.

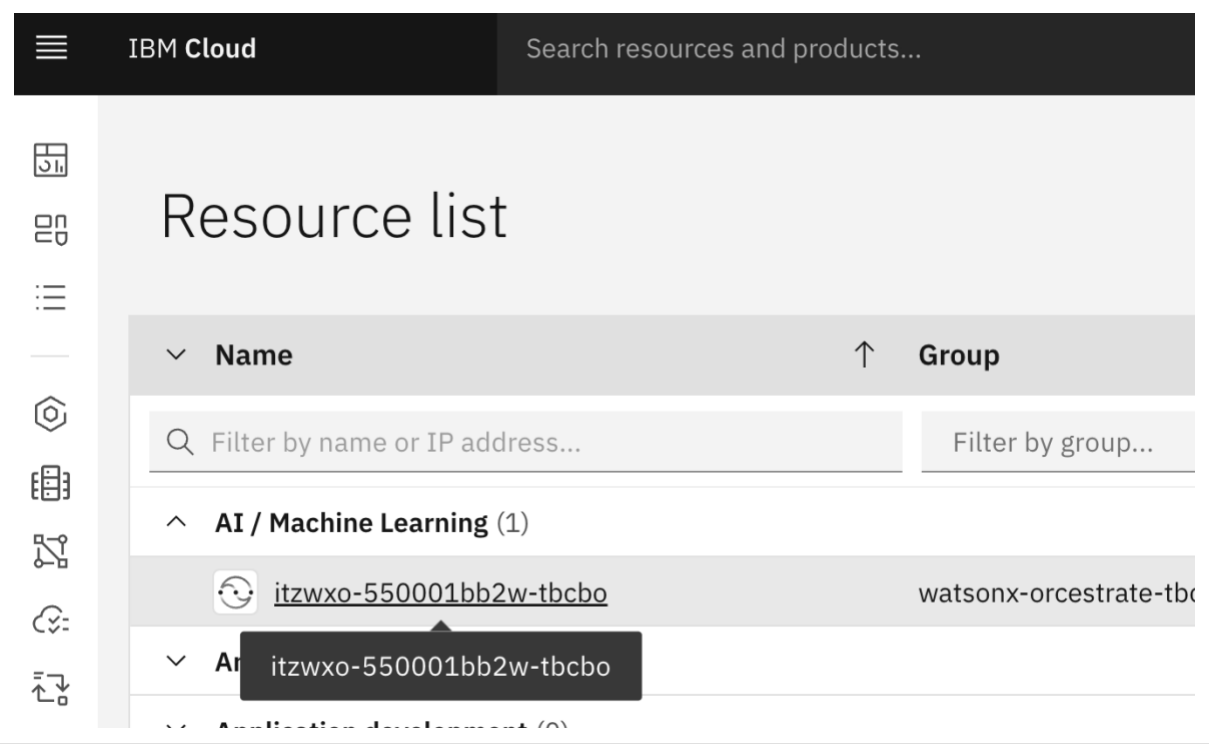
3.5 Open watsonx Orchestrate

1 From the dashboard, open the navigation menu at the top left and choose **Resource list**.



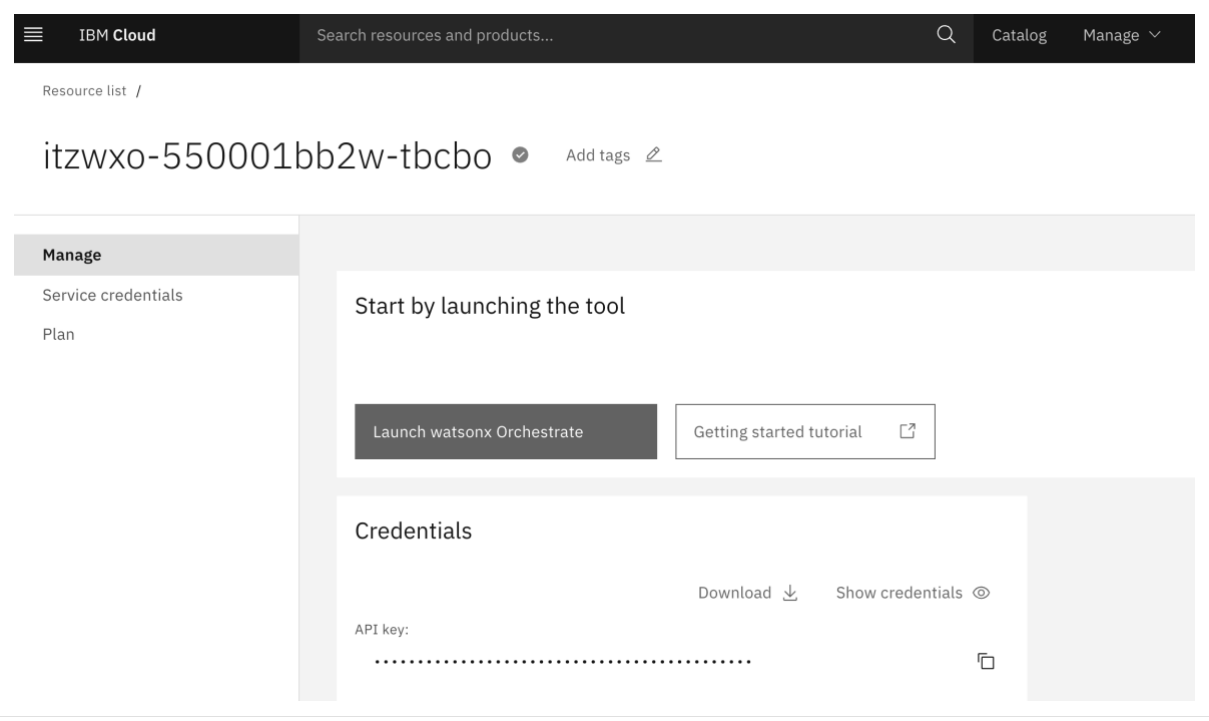
The main menu. Resource list is the third entry.

2 Expand **AI / Machine Learning** and select the **watsonx-Hackathon Orchestrate** service.



The watsonx Orchestrate service under AI / Machine Learning. In your hackathon account the instance name will differ from the one shown here.

3 On the service page, **copy the API key and keep it somewhere safe** — you will need it to connect Bob — then select **Launch watsonx Orchestrate**.

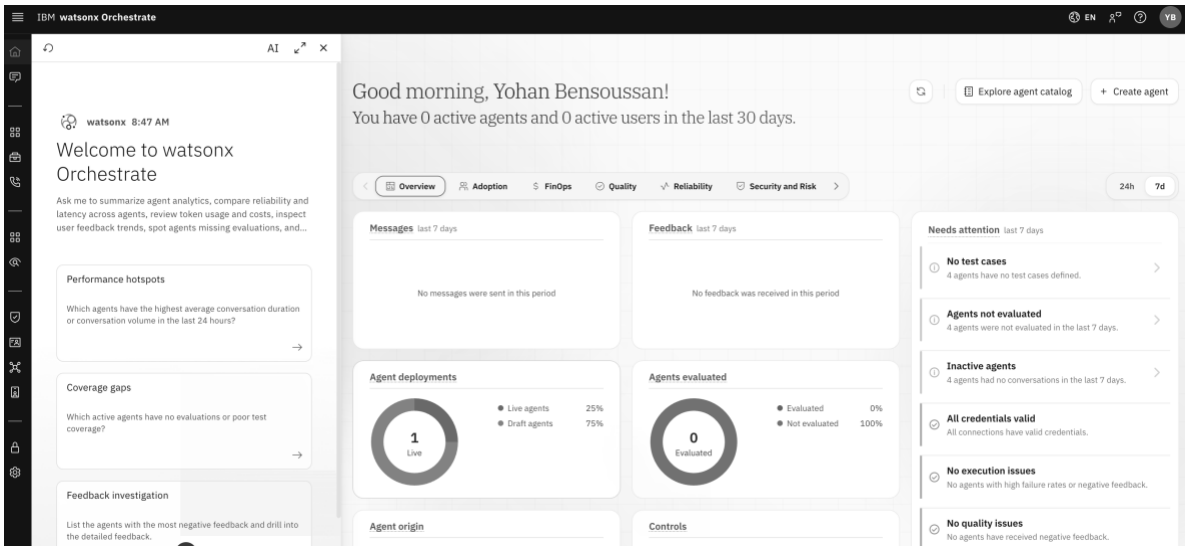


Launch watsonx Orchestrate starts the platform. The API key is below, under Credentials — use Show credentials or the copy icon.

IMPORTANT

Copy the API key before you launch. Going back for it afterwards is a detour you do not need.

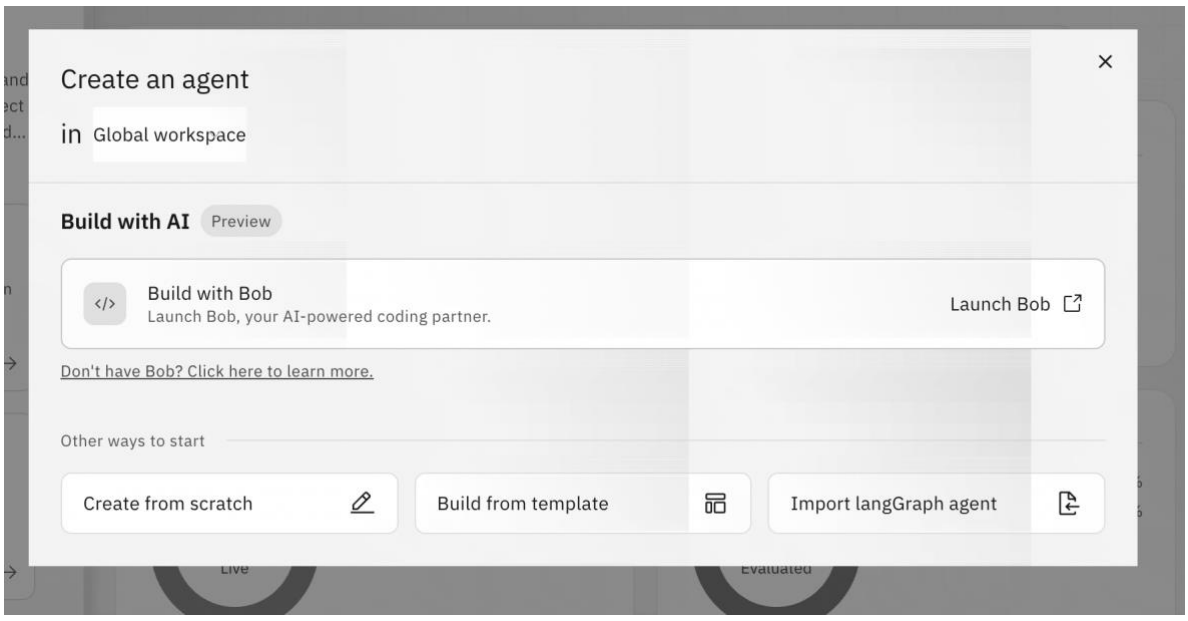
4 You arrive in watsonx Orchestrate, with a welcome message and a new chat window.



You are in watsonx Orchestrate. Create agent is at the top right.

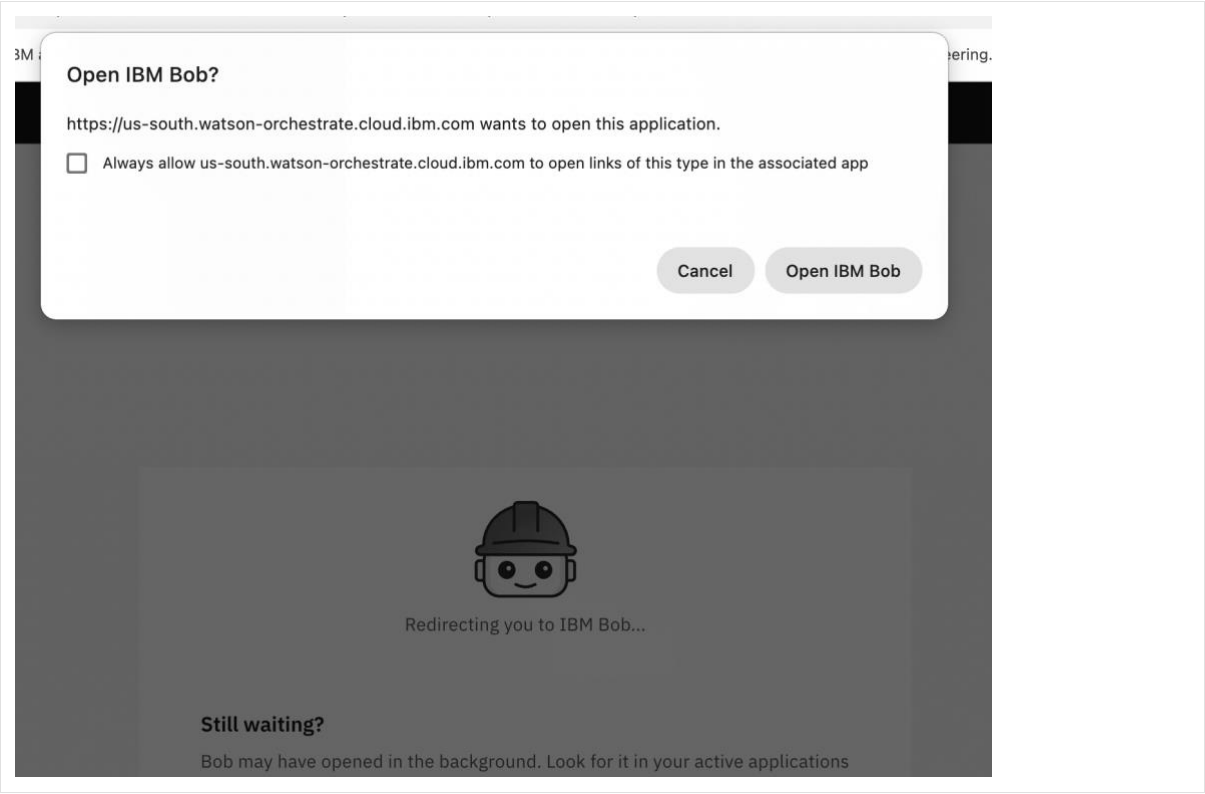
3.6 Create an agent with Bob

1 Click **Create agent**. The dialog offers **Build with Bob** — choose **Launch Bob**.



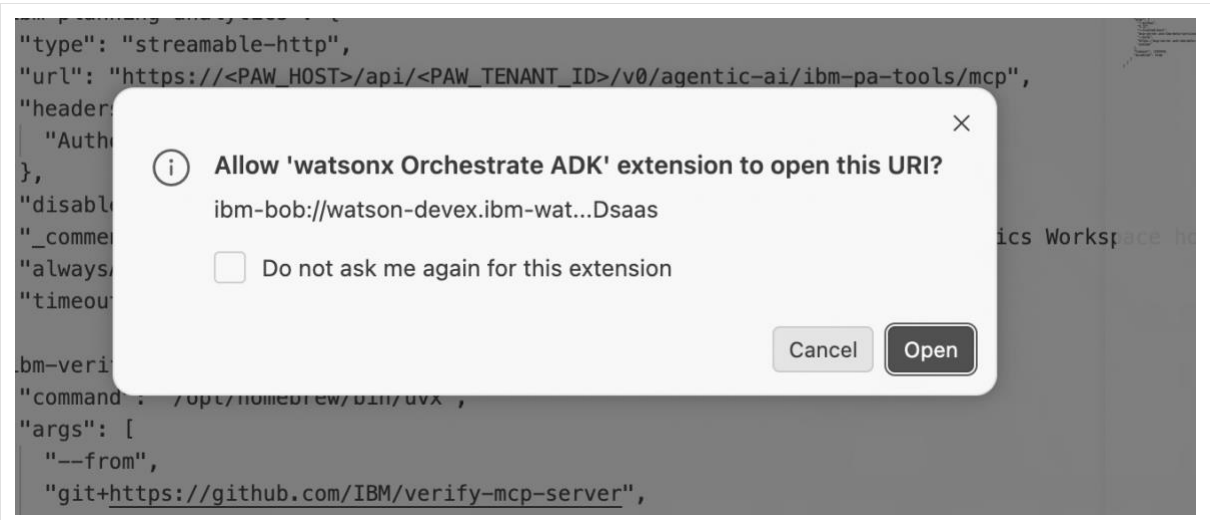
Create an agent → Build with Bob → Launch Bob.

2 Your browser asks permission to open Bob. Allow it.



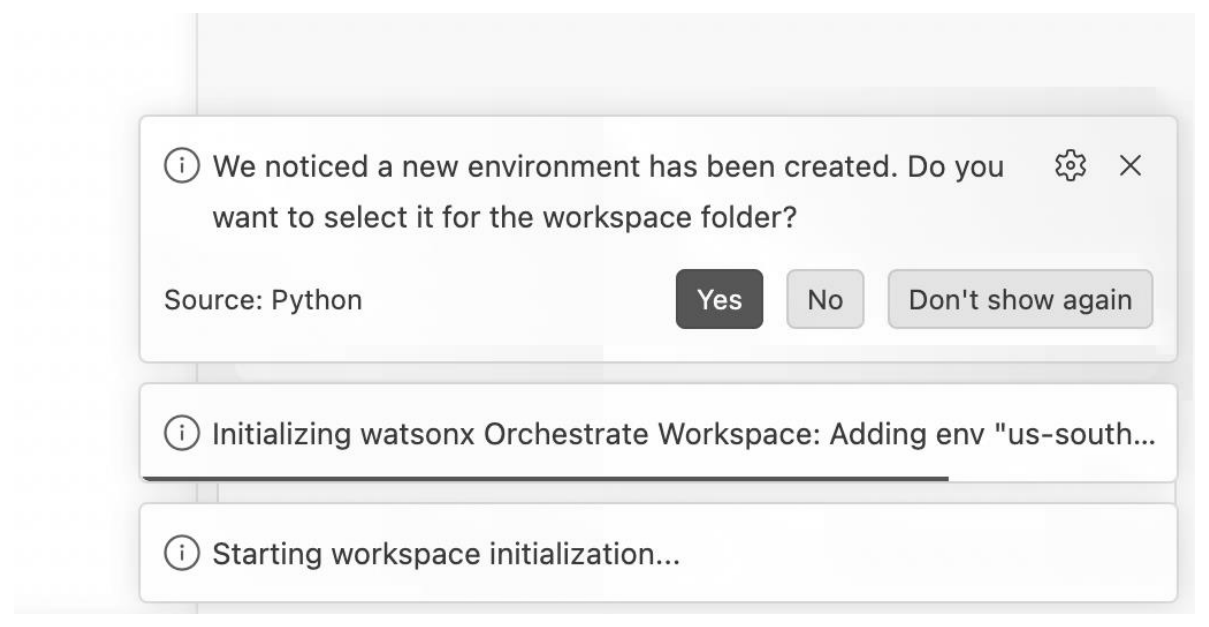
The browser hands off to the Bob application.

3 Bob asks to allow the **watsonx Orchestrate ADK** extension to open the URI. Choose Open.



Allow the watsonx Orchestrate ADK extension.

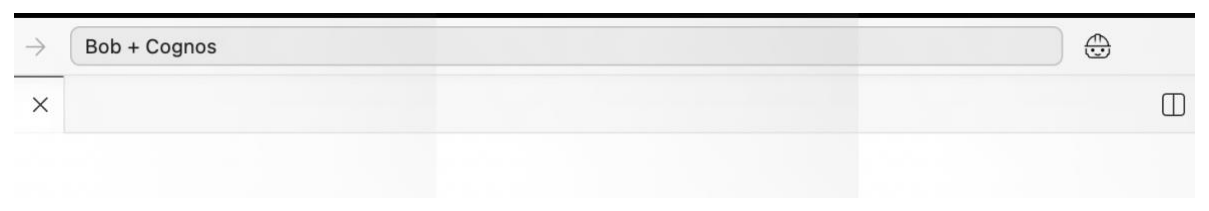
4 The first time only, Bob installs what it needs to run watsonx Orchestrate and initialises the workspace. Let it finish.



First-run setup: Bob creates the environment and initialises the workspace.

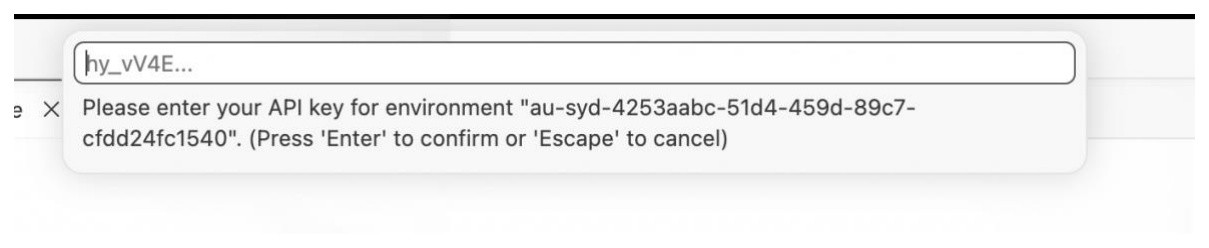
3.7 Connect Bob to your environment

1 Bob asks for your API key, in the search field in the header.



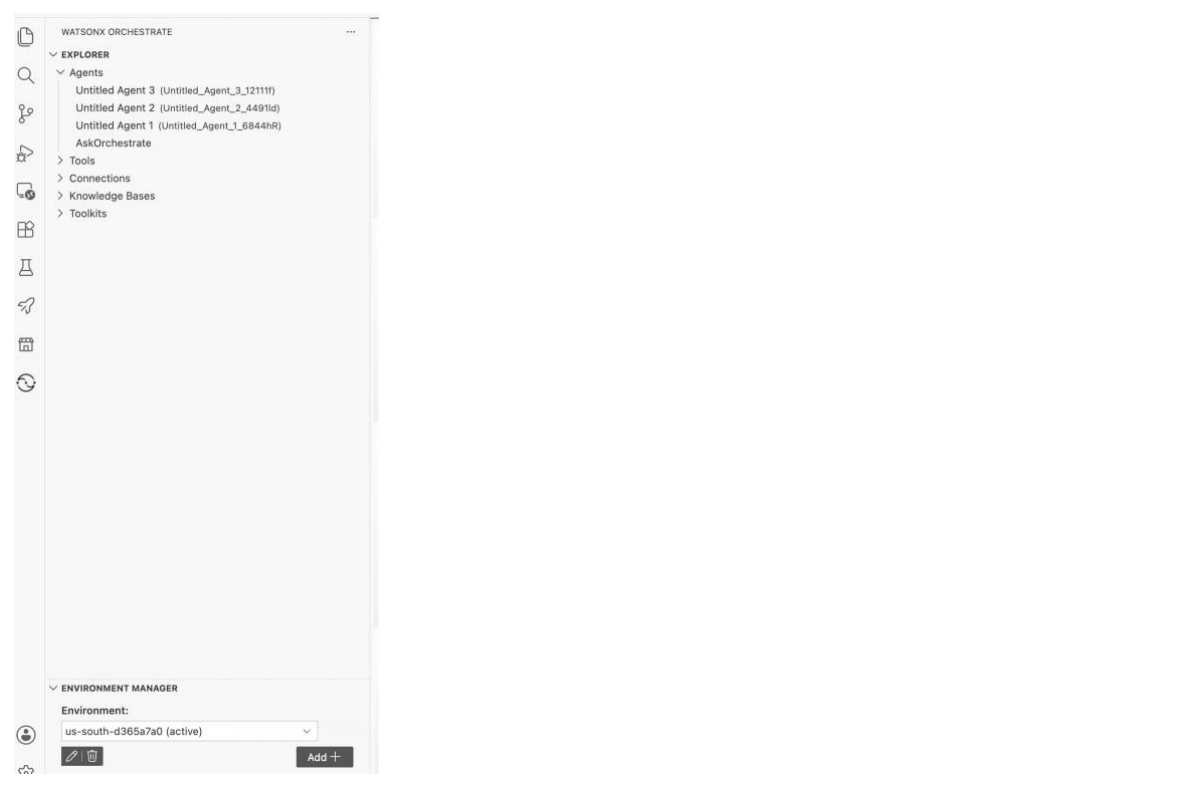
The header field is where the prompt appears.

2 Paste the API key you copied in 3.5 and press **Enter**.



Paste the key for the environment shown, then press Enter.

3 A **watsonx Orchestrate** panel appears on the left, with everything on the platform connected — agents, tools, connections, knowledge bases and toolkits.

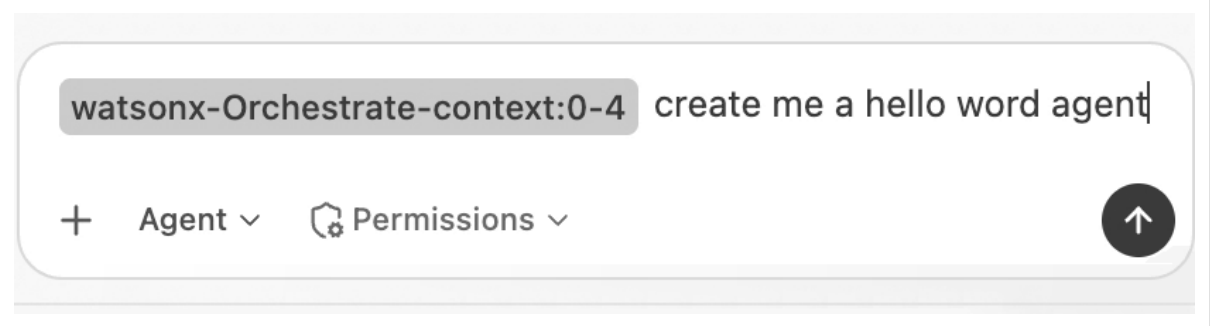


The watsonx Orchestrate panel inside Bob. The Environment Manager at the bottom shows which environment is active.

That is the whole connection. You can start building.

3.8 Build your first agent

Ask Bob in plain language. Here is a hello world agent, start to finish.



Ask for what you want. The watsonx Orchestrate context is attached to the prompt automatically.

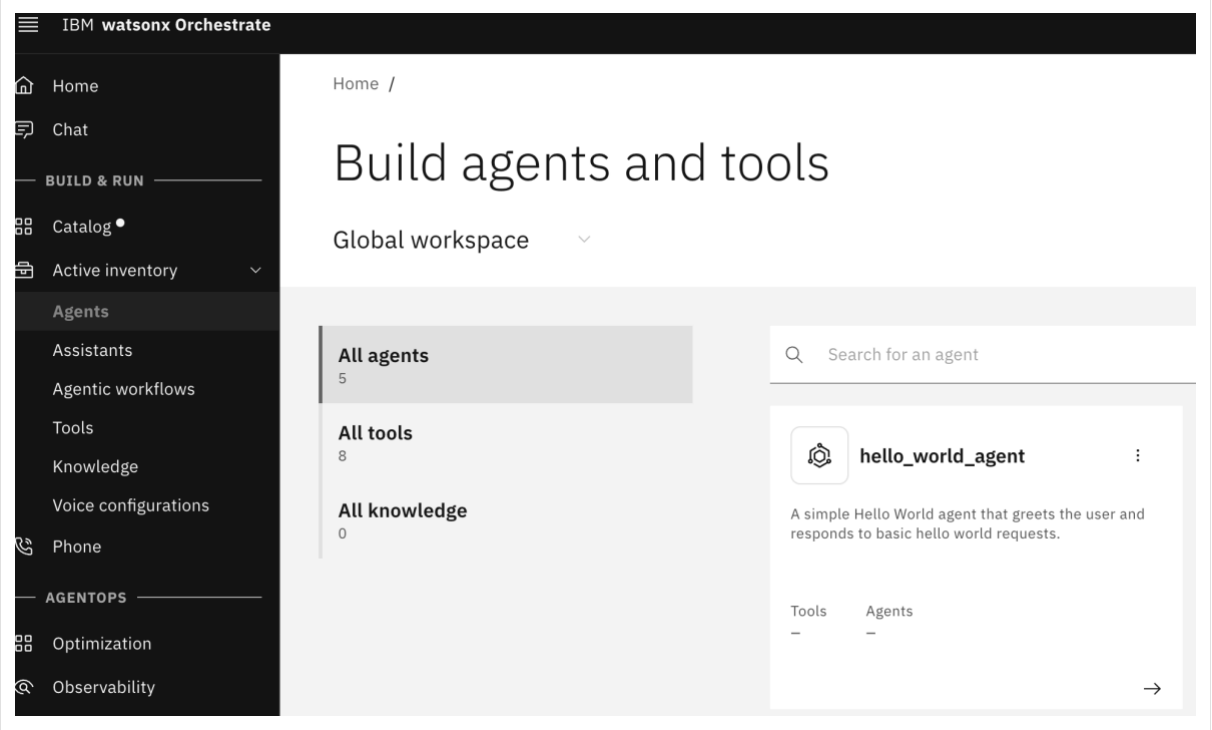
When Bob has finished, the new agent appears in the panel on the left.



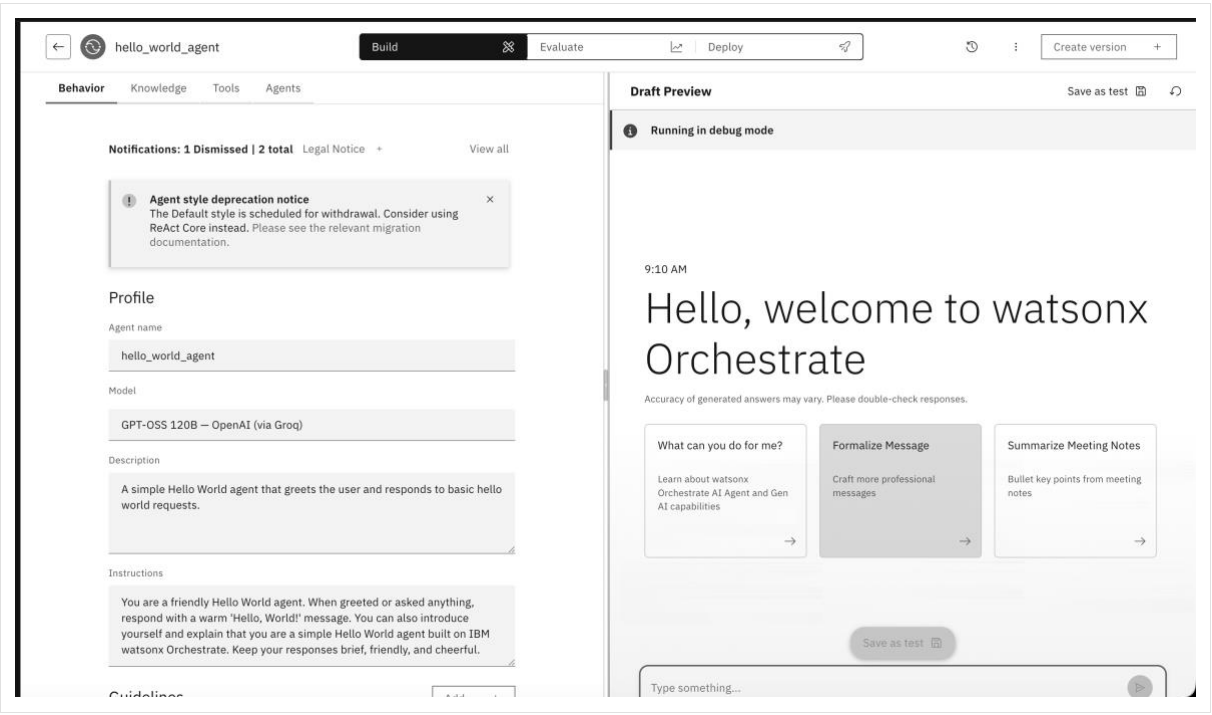
hello_world_agent, created by Bob, now listed among the agents.

3.9 See it in the platform

Go back to watsonx Orchestrate in the browser. What Bob built is there, ready to open and test.



The agent appears in Build agents and tools, in the platform.



Open it to see the configuration Bob generated — and use Draft Preview on the right to talk to it.

4 Next steps, learning resources and submission requirements

You now know how to get onto your hackathon accounts, run Bob, connect watsonx Orchestrate, and build an agent. What sits on top of that is considerably larger.

- Create agents, and the tools they use
- Connect agents to each other and orchestrate them
- Build tools from OpenAPI, MCP servers or agentic workflows
- Test, evaluate and deploy what you build

Both products ship a large set of hands-on exercises — including several that use Bob and watsonx Orchestrate together, such as building agents and MCP tools with Bob, and turning BPMN diagrams into deployable agents. They are the fastest way to find capabilities you would not otherwise discover.

> **TIP**
> **Start with the education before you start building.** Bob: https://bob.ibm.com/docs/ide · watsonx Orchestrate: see the platform's own learning material.

> **IMPORTANT**
> **Use the hackathon-provisioned accounts for every exercise you try.** Running them on a personal Bob account spends your own allowance and produces nothing the judges can see.

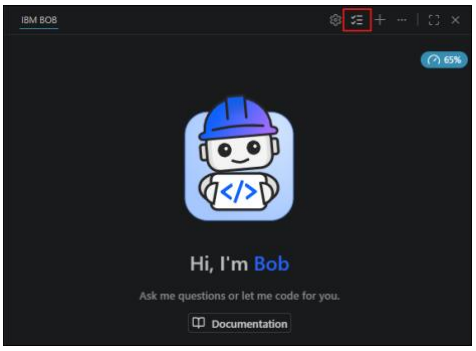
Before submitting your project, complete the following required Bob task session summary process.

4.1 Upload Bob task session summary

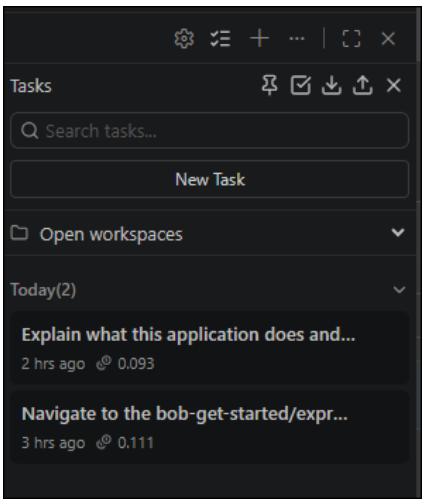
Each participant must upload all relevant Bob task session summary screenshots as part of your project submission. These screenshots must be included in your **final code repository** as a **submission deliverable**.

Follow the steps below to complete this process:

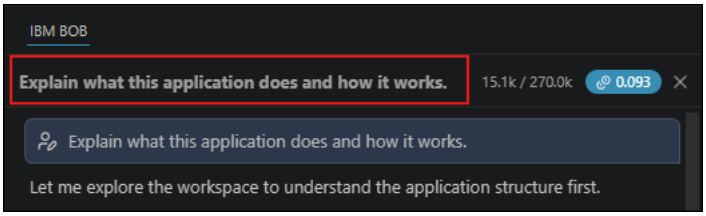
- Create a folder named **bob_sessions** in your project submission code repository.
- In your Bob IDE's chat interface, select **Tasks** to display the task list.



- Select a task from the list related to your project submission and the selected task will open in the chat panel. Confirm that you are in the correct project workspace. If your submission includes tasks from multiple workspaces, select **All** to view tasks across all relevant workspaces.

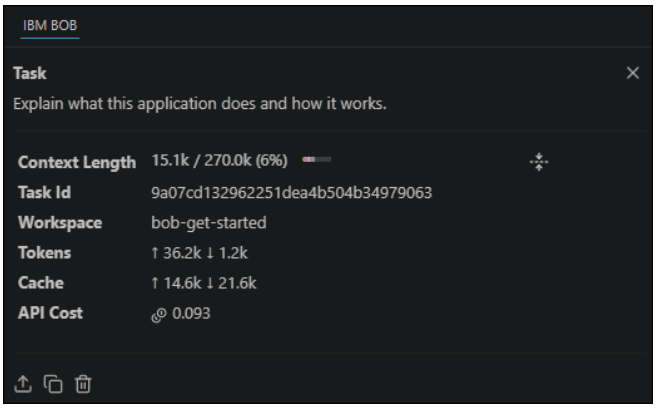


- Select the task header. A task session consumption summary will be displayed.



- Take a screenshot of the task session consumption summary. Save screenshots in **PNG** format where possible for better text clarity. Use a clear file name that includes your team name, task number, and short task description.

  Example: teamalpha_task01_login_flow_summary.png



- Now, repeat the steps above for all the tasks related to your project submission.
- Upload all task session consumption summary screenshots to the **bob_sessions** folder in your code repository.

Important: Before submitting your project, ensure that your team final code repository includes the **bob_sessions** folder with all required screenshots relevant to the project solution.

Good luck.

And remember — an IBM engineering team is on call for the whole challenge. Use them early rather than late.